Robotic arm segment board v1

Power: inputs, protection, eFuses, 3.3 V

File: power.kicad_sch

ESP32-S3 controller + USB-C

File: mcu.kicad_sch

I2C peripherals: master select, PCA9685, TCA9548A, ADS1115

File: i2c.kicad_sch

Arm connectors + hardware connection LEDs

File: connectors.kicad_sch

Raspberry Pi header + RS-485 LINK

File: pi_link.kicad_sch

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Sheet: /

File: segment_board.kicad_sch

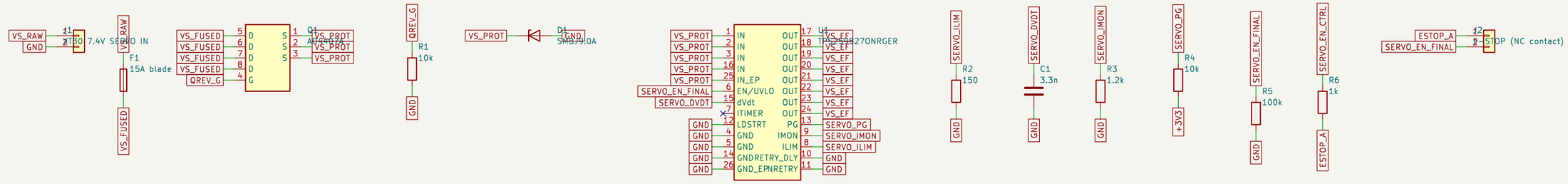
Title: Robotic arm segment board v1

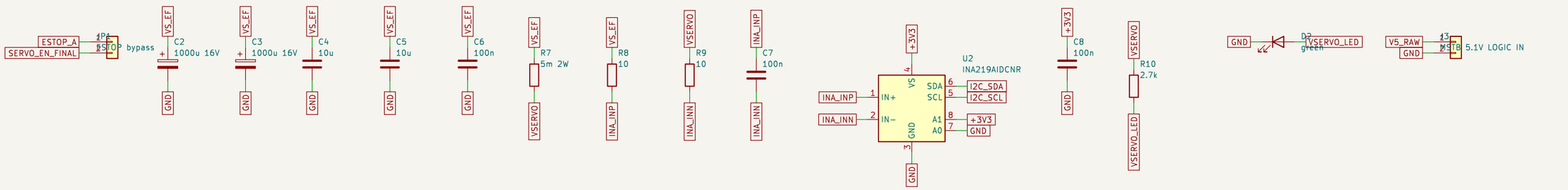
Size: A4	Date: 2026-11-08	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/6

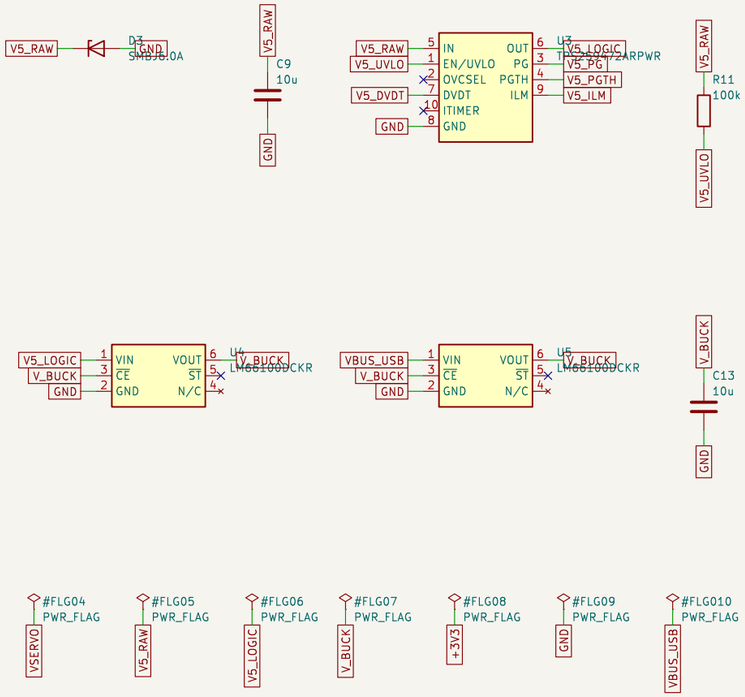
7.4 V SERVO: XT30 -> 15 A fuse -> P-MOSFET reverse polarity -> TVS -> eFuse TPS259827 (ILIM ~10 A, soft-start, EN) -> bulk caps -> INA219 shunt -> VSERVO

5.1 V LOGIC: MSTB -> TVS -> eFuse TPS259472A (ILM ~4.4 A, reverse blocking, OV clamp 5.7 V) -> V5_LOGIC (Pi 5V pins, 3.3 V buck)

E-STOP (board level): NC contact in series with the eFuse EN line; central E-stop cuts the 7.4 V line at the base (ADR-0015)







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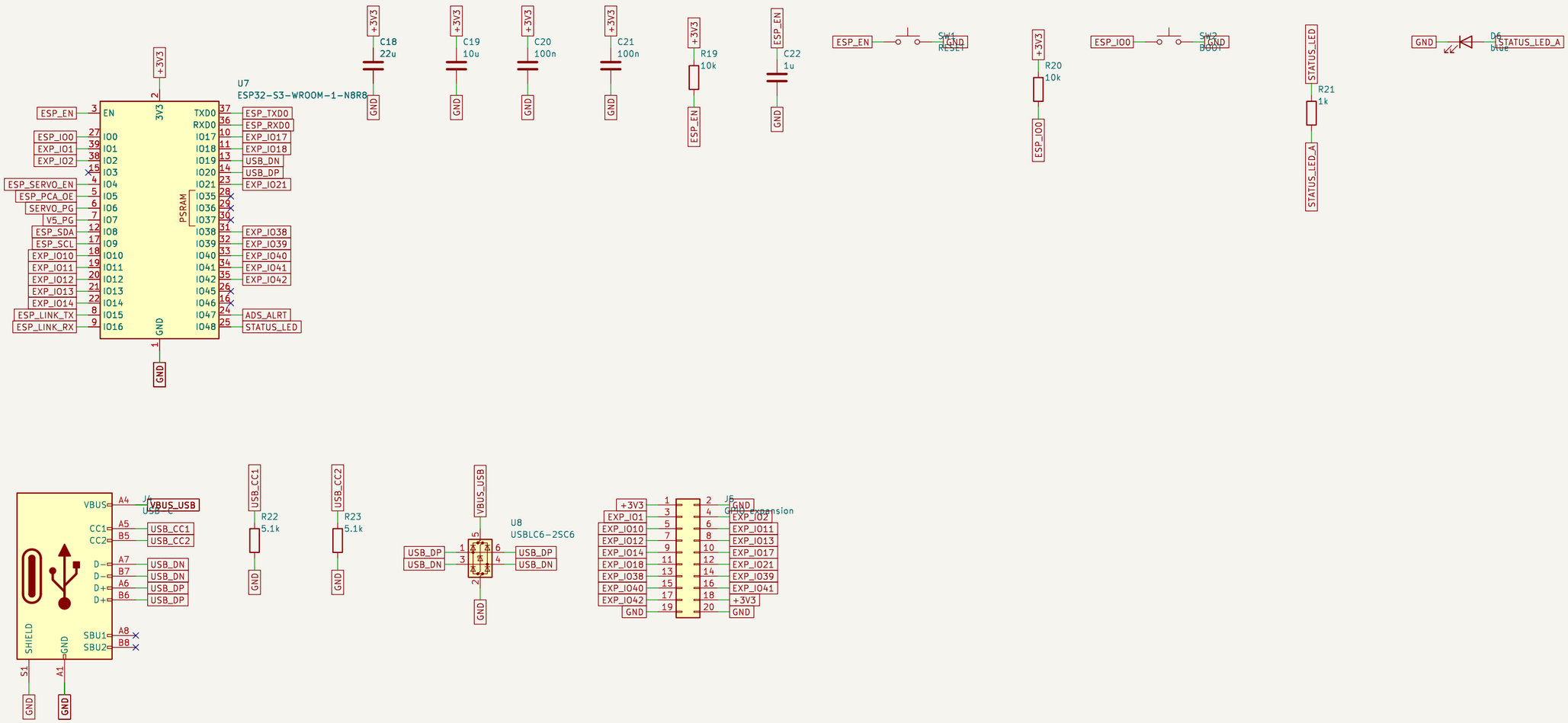
Sheet: /Power: inputs, protection, eFuses, 3.3 V/

File: power.kicad_sch

Title: Power: inputs, protection, eFuses, 3.3 V

Size: A3	Date: 2026-11-08	**Rev: v1**
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ESP32-S3-WROOM-1-N8R8. Antenna at board edge, no copper under it. IO35-37 used by octal PSRAM; IO0/3/45/46 strapping.



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Sheet: /ESP32-S3 controller + USB-C/

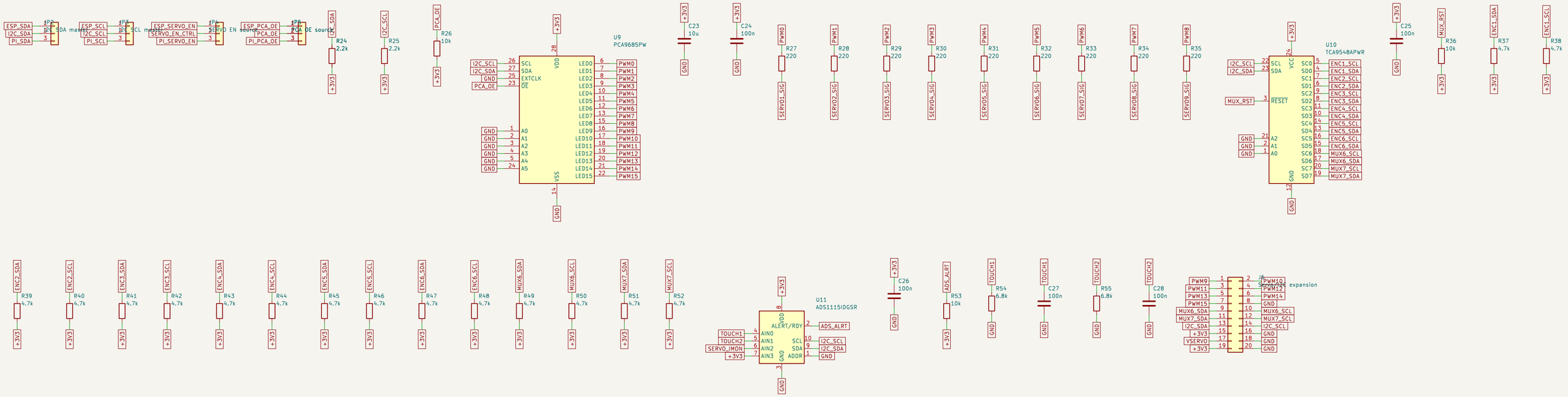
File: mcu.kicad_sch

Title: ESP32-S3 controller + USB-C

Size: A3	Date: 2026-11-08	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/6

JP: shunt 1–2 = ESP32 is I2C master / control; shunt 2–3 = Raspberry Pi. Never both.

Addresses: PCA9685 0x40, INA219 0x44, ADS1115 0x48, TCA9548A 0x70, AS5600 0x36 behind the mux.



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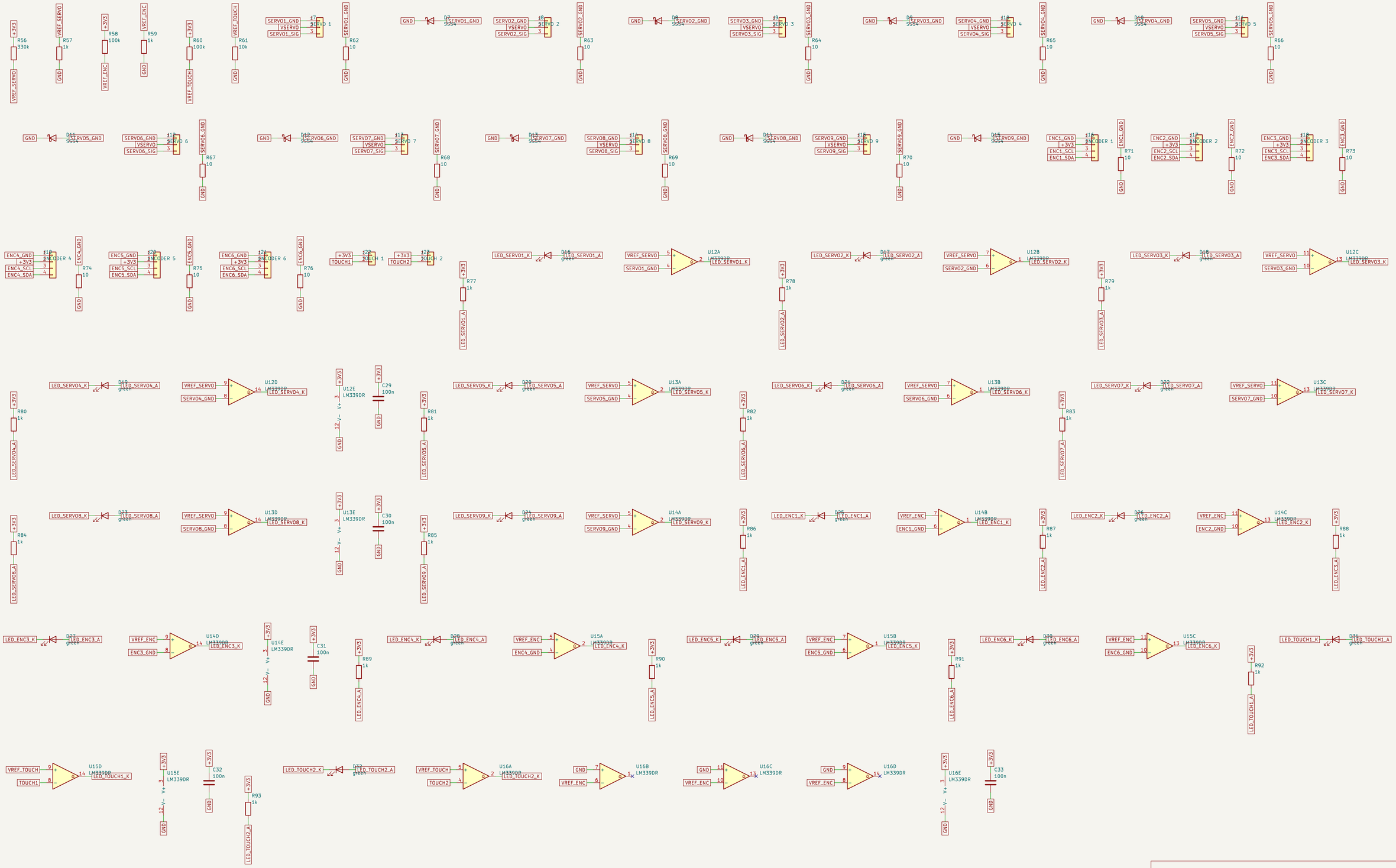
Sheet: /I2C peripherals: master select, PCA9685, TCA9548A, ADS1115/

File: i2c.kicad_sch

Title: I2C peripherals: master select, PCA9685, TCA9548A, ADS1115

Size: A2	Date: 2026-11-08	**Rev: v1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 4/6

Connection LED = current detection on each connector's ground return (no extra pins, no software).
Comparator LM339 (open collector): output low (LED on) when the sense voltage exceeds the reference.
SERVO sense 10 R + SS54, VREF ~10 mV -> detect at 0.5–1.5 mA (servo idle 4–20 mA typ.; confirm by measurement).



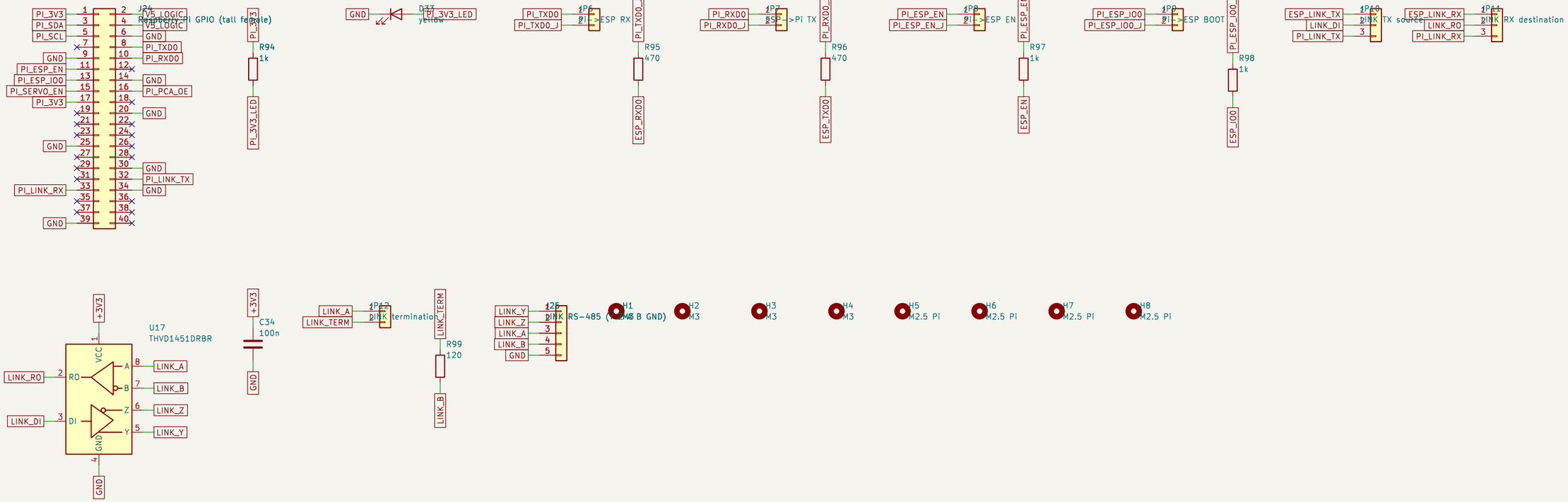
Robotic arm segment board v1
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Sheet: /Arm connectors + hardware connection LEDs/
File: connectors.kicad_sch

Title: Arm connectors + hardware connection LEDs

Size: A2	Date: 2026-11-08	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/6

Pi mounts upside down on top: footprint must be MIRRORED relative to a standard HAT (see SPEC R3).

Pi powered from its 5V pins through the 5.1 V eFuse. PI_3V3 is the Pi's own 3.3 V (LED only, never tied to +3V3).



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Sheet: /Raspberry Pi header + RS-485 LINK/

File: pi_link.kicad_sch

Title: Raspberry Pi header + RS-485 LINK

Size: A3	Date: 2026-11-08	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/6